

Shakespeare with Ladders: Composition, Derived Power, and the Second Refusal of a Receiver Language

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Abstract

Shakespeare is a receiver language whose single primitive is the cut: a commitment that fixes boundary in a finite weighted contact graph, deposits residue at least the floor $\beta > 0$, and advances a monotone clock. The language has one composition rule—residue chaining—which sums the boundary a process commits. Summation is order-insensitive, and the language’s own reversal lemma turns that into a limitation: no predicate definable from chained residue can orient a process, and none can say how much of a process a given step accomplishes.

We add a second composition rule. A *rung* carries one number, its *power* $\pi \in [0, 1]$: the fraction of the remaining above-floor gap it closes. A *ladder* composes rungs multiplicatively, $1 - \prod_i (1 - \pi_i)$. Where residue chaining is extensive and additive, power is intensive and multiplicative, and the two answer different questions about the same chain.

Three results are new. First, power need not be declared: we derive it from the separation cost against the medium, computed on a ball of declared radius, and we show that *radius is a resolution parameter, not a threshold*. Second, intensivity is a condition rather than a consequence—and it is delicate. Deriving power from a local cut key is not sufficient; the normaliser must be local too. Using the global floor breaks intensivity, and we found this because the test was built to fail.

We report the resulting order-dependence as *graded*: intensivity reduces it by an order of magnitude and monotonically in locality, but does not abolish it under mutation. Third, the ladder supplies the language’s second refusal, discharging the standing requirement that a construct which never refuses classifies nothing.

We give the grammar, four typing rules, four reduction rules, and proofs of progress, preservation, and clock monotonicity for the extended language. A free operation is admitted only under the argument the language already uses for its one free observation—that it individuates nothing new—and we amend the two uniqueness claims that the extension falsifies rather than leaving them to be discovered. Twenty-three checks run, of which twenty-one are scored and all twenty-one pass; four are negative controls, one is reported non-discriminating with its reason, and one records a claim the suite cannot test.

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II	Rungs and Ladders	6	1.1	What the language already does, and the one thing it cannot say	
4	Derived Power	6		Shakespeare is a language in which evaluation is measurement [Sachikonye, 2026b]. Its state is a finite weighted contact graph $\Gamma = (V, E, w)$ with a distinguished medium vertex $\mathbf{m}$ adjacent to every other vertex. Its single primitive is the <i>cut</i> : a commitment that fixes a set of edges as realised separations, deposits residue $\rho = w(\text{cut}(S)) \geq \beta$, and advances a monotone counter M . Every surface verb— <i>fold</i> , $\sim$, <i>close</i> , <i>track</i> , <i>catalyze</i> , <i>mutate</i> , <i>complete</i> —is the cut at some arity.	
4.1	Intensivity	7		Composition is residue chaining: a chain $C = (S_1, \dots, S_k)$ commits total boundary $\sum_i \rho_i$ and k cut events. This is the language's one composition rule, and it is <i>extensive</i> : it adds, it grows with the length of the chain, and it is invariant under reordering. That last property is not an accident but a theorem—the reversal lemma [Sachikonye, 2026a]—and the language draws a sharp negative consequence from it: no predicate definable from the residues, the cut count, or the committed boundary of a chain distinguishes a chain from its reversal. The medium paper answers that limitation for <i>orientation</i> by going outside the chain, to the medium.	
5	Composition	8		There is a second question the chain cannot answer, and it is not about direction. Given a process that reaches its target, which step accomplished most of the reaching? Total committed boundary does not say: a step may commit a large boundary and move the process barely at all, and the sum records only that the boundary was committed. Sensitivity, control, and the notion of a rate-limiting step are all outside what an extensive aggregate can express.	
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the specification claims there is exactly one (theorem 1.1).

A *rung* carries exactly one datum: its *power* $\pi \in [0, 1]$, the fraction of the remaining above-floor gap that it closes. A *ladder* is a finite sequence of rungs, and its composite power is

$$\pi(\gamma_1 \cdots \gamma_n) = 1 - \prod_{i=1}^n (1 - \pi_i). \quad (1)$$

This is *intensive* where residue chaining is extensive: it is a fraction rather than a total, it is bounded by 1 however long the ladder, and adding a rung of power 0 changes nothing. The two rules coexist because they answer different questions about the same chain—how much boundary was committed, and how much of the distance was closed—and section 6 shows that neither determines the other.

Three contributions are specific to this paper.

Power is derived, not declared. The prior formulation took powers as measured inputs and said so as a limitation. Here a rung’s power is computed from the *separation cost* $\sigma(v)$ —the minimum weight of a cut placing v on one side and the medium on the other—evaluated on a ball of declared radius ρ around v . The radius is a *resolution parameter*, not a *threshold on a score*: raising it admits more of the graph into the computation, so a coarse ladder matches more permissively and a fine one discriminates more. Section 4 gives the construction and section 12 the measured radius sweep.

Intensivity is a condition, and it is delicate. If power depended on what else was in the graph, composition would depend on the order in which rungs were derived and eq. (1) would be the wrong law. We state intensivity as a requirement and test it. It fails for the obvious construction: a local cut key normalised by the *global* floor is not intensive, because β is a minimum over every edge and one distant small edge moves it everywhere. Both the cut key and its normaliser must be local (theorem 4.7). We report this rather than presenting only the construction that

works, because the near-miss is what localises the failure, and because the test was built so that it could fail.

The honest form of the consequence is graded. Under sequential derivation—in which each rung sees the graph its predecessors have modified—intensivity reduces order-dependence by an order of magnitude and monotonically in locality, but does not abolish it (theorem 5.8). We state the graded result rather than the clean one because the clean one is false.

The ladder refuses. The language holds that a representation earns a positive result only if some presentable object would have received a negative one [Sachikonye, 2026a]. A construct that always succeeds classifies nothing. The ladder refuses a declared target its declared rungs cannot reach, and the refusal is *tight*: it rejects no ladder that could have succeeded (theorem 10.5). Section 10 shows the refusal is non-vacuous by exhibiting presentable ladders that trigger it.

1.3 What we do not claim

We do not claim the ladder is better than residue chaining; it answers a different question, and section 6 is careful that neither aggregate determines the other. We do not derive powers from chemical structure—the derivation is from the contact graph, which is itself a model. We do not claim exact order-independence, and theorem 5.8 says what we measured instead. We do not claim the extension is conservative in the strong sense that no existing program changes meaning; it is conservative in the weaker and checkable sense that no existing rule is modified and the added rules apply only to added syntax (theorem 9.12).

1.4 The two claims this paper falsifies

An extension that quietly contradicts its host is worse than one that does so openly. Two published claims are falsified by what follows, and we amend both in place rather than leaving a reader to find them.

Remark 1.1 (Amendment: one composition rule becomes two). The specification states that Shakespeare has “one verb (the cut) and one composition rule (residue chaining)” [Sachikonye, 2026b]. With the ladder this is false as written. The verb claim survives—a rung commits a cut and nothing else—but the composition claim must read: *one verb, and two composition rules over its chains, one extensive and one intensive*. Section 6 shows the two are independent, so the amendment is not a restatement.

Remark 1.2 (Amendment: the free observation is a class, not an instance). The medium paper states of role observation that “this is the one Shakespeare observation that is free, and it is free because it individuates nothing new” [Sachikonye, 2026a]. The reason is correct and we adopt it verbatim; the uniqueness is not. Reading a ladder’s composite power is free for exactly the same reason. The claim should read: *an observation is free precisely when it individuates nothing new, and role observation was the first such*. Theorem 9.5 makes the criterion a theorem rather than a case-by-case licence, which is what turns the amendment into a strengthening.

Part I

The Substrate

2 Graph, Floor, Separation Cost

We restate what the extension builds on. Nothing in this section is new; it is here so that the paper is self-contained and so that the exact form of each quantity is fixed before it is used.

Definition 2.1 (Contact graph). A *contact graph* is a finite weighted graph $\Gamma = (V, E, w)$ with a distinguished *medium* vertex $\mathbf{m} \in V$ adjacent to every other vertex, and $w : E \rightarrow \mathbb{R}_{>0}$ strictly positive. Vertices $v \neq \mathbf{m}$ are *items*; an edge between two items is a *contact*; $w(e)$ is the cost of the separation it maintains. We write $\Omega = \sum_{e \in E} w(e)$.

Definition 2.2 (Cut, residue, floor). For $S \subseteq V$ the *cut* is $\text{cut}(S) = \{ \{u, v\} \in E : |\{u, v\} \cap S| = 1 \}$ with weight $w(\text{cut}(S)) = \sum_{e \in \text{cut}(S)} w(e)$. A set S is *admissible* if it contains at least one item and omits $\mathbf{m}$. The *floor* is $\beta = \min_{e \in E} w(e)$.

Lemma 2.3 (Positive floor). $\beta > 0$, and $w(\text{cut}(S)) \geq \beta$ for every admissible S .

Proof. E is finite and non-empty and every weight is strictly positive, so the minimum is attained at a strictly positive value. An admissible S contains some item v and omits $\mathbf{m}$, so $\{v, \mathbf{m}\} \in \text{cut}(S)$ and the cut weight is at least $w(\{v, \mathbf{m}\}) \geq \beta$. $\square$

Remark 2.4 (Finiteness does the work). Theorem 2.3 uses finiteness twice and cannot be repaired without it. An infinite edge set of strictly positive weights may have infimum zero: the weights $1/n$ are each positive and admit no positive lower bound. The floor is a property of a finite system, and every result below inherits that hypothesis.

Definition 2.5 (Separation cost). For an item v the *separation cost* is

$$\sigma(v) = \min_{S \ni v, \mathbf{m} \notin S} w(\text{cut}(S)), \quad (2)$$

the minimum weight of a cut placing v on one side and the medium on the other. The minimum is over a finite non-empty family, hence attained, and equals a maximum flow [Ford and Fulkerson, 1956].

By theorem 2.3, $\sigma(v) \geq \beta > 0$ for every item.

Remark 2.6 (Why the medium is the reference). Equation (2) measures v against the medium and not against another item. This is forced: the medium is by construction what has no surroundings within Γ , so it is the only vertex against which a separation cost is well posed without a further choice. Measuring v against an arbitrary item u would make the quantity depend on u , and a power so defined would not be a property of v .

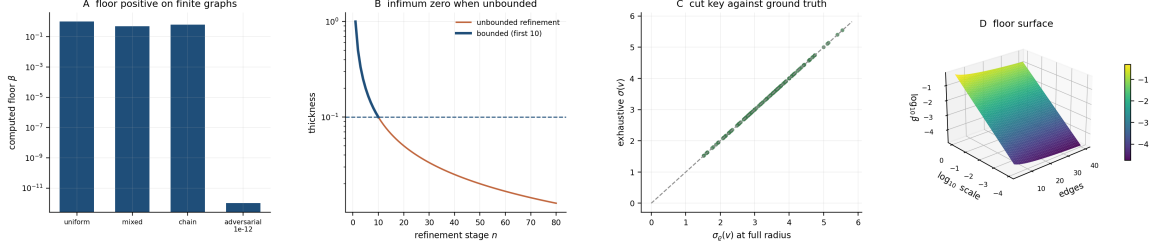


Figure 1: **The substrate: a floor that is computed, the hypothesis it requires, and the cut key against ground truth.** (A) Floor computed—not assumed—as the minimum edge weight over four finite contact graphs, including an adversarial instance whose smallest weight is 10^{-12} . Every instance returns a strictly positive floor, because positivity follows from finiteness together with strict positivity of the weights and not from their scale. (B) The failure branch. A system permitted to refine without limit generates thicknesses $1/n$ (orange): each is strictly positive and the infimum is zero, with every ε down to 10^{-9} eventually cleared. Truncating the same sequence to a finite system (blue, first ten stages) restores a positive floor at 0.1 (dashed). The uniform floor therefore requires finiteness, and the chart is included so that the requirement is visible rather than tacit. (C) The local separation cost of eq. (3), evaluated at a radius covering the graph, against an independent exhaustive enumeration of every bipartition, on 160 graphs. Points lie on the identity to machine precision. This checks the implementation against ground truth rather than restating the definition. (D) Floor surface over edge count and weight scale. The surface is positive everywhere on the domain and descends toward zero only as the scale does; increasing the edge count at fixed scale lowers the floor without abolishing it, which is the content of a minimum over a finite set.

3 Resolution: the Ball and the Local Floor

The separation cost of eq. (2) ranges over every subset of the graph. We want a quantity that can be computed from a bounded neighbourhood, both because that is what makes it intensive (section 4) and because the bound is itself a meaningful parameter.

Definition 3.1 (Ball). The *ball* of radius $\varrho \in \mathbb{N}$ about an item v is the set $B_\varrho(v)$ of items reachable from v by at most ϱ contacts, where the medium is not traversed. Thus $B_0(v) = \{v\}$, and $B_{\varrho+1}(v)$ adds the item neighbours of $B_\varrho(v)$.

Remark 3.2 (The medium is not traversed, and why). Excluding $\mathbf{m}$ from traversal is not a convenience. The medium is adjacent to every item, so a path through it joins any two items in two steps and every ball of radius 2 would be the entire graph. The resolution parameter would then have exactly two settings and would not be a resolution parameter.

Items are near each other when they share contacts, not when they share the medium.

Definition 3.3 (Local separation cost and local floor). The *local separation cost* at radius ϱ is

$$\sigma_\varrho(v) = \min_{v \in S \subseteq B_\varrho(v)} w(\text{cut}(S)), \quad (3)$$

the minimum taken over subsets of the ball, with the cut still evaluated in the whole graph. The *local floor* is $\beta_\varrho(v) = \min\{w(e) : e \subseteq B_\varrho(v) \cup \{\mathbf{m}\}\}$, the least edge weight among edges internal to the ball together with its medium edges.

Lemma 3.4 (The local floor is a floor). $0 < \beta \leq \beta_\varrho(v) \leq \sigma_\varrho(v)$ for every item v and radius ϱ .

Proof. $\beta_\varrho(v)$ is a minimum over a non-empty subfamily of E —non-empty because $\{v, \mathbf{m}\}$ is always in it—so it is attained, is at least the minimum over all of E , which is $\beta > 0$, and the first two inequalities follow. For the third, the minimising S in eq. (3) contains v and omits $\mathbf{m}$,

so $\{v, \mathbf{m}\} \in \text{cut}(S)$ and $\sigma_\varrho(v) \geq w(\{v, \mathbf{m}\}) \geq \beta_\varrho(v)$, the last step because $\{v, \mathbf{m}\}$ is one of the edges over which $\beta_\varrho(v)$ minimises. $\square$

Principle 3.5 (Resolution, not threshold). The radius is a *resolution parameter*, not a threshold on a score. It does not set a level above which a quantity counts; it sets how much of the graph enters the computation at all. Its effect is therefore monotone and stateable in advance, and there is no setting at which it is uniformly best—only settings appropriate to a question.

Proposition 3.6 (The direction of the sweep, and its mechanism). *Raising the radius makes the derived power identify more pairs of items, not fewer. The mechanism is ball overlap: at radius 0 the balls are disjoint singletons and each item minimises over its own incident edges, so distinct items generically receive distinct powers; as the radius grows the balls overlap and eventually coincide, so distinct items minimise over nearly the same edge set and their powers converge.*

Proof. Ball overlap is non-decreasing in the radius, since $B_\varrho(v) \subseteq B_{\varrho+1}(v)$ for every v , and on a connected item graph of diameter d every ball of radius at least d is the whole item set, at which point σ_ϱ and β_ϱ are the same function for every item and all powers coincide. At radius 0, $B_0(v) = \{v\}$ and the minimisation in eq. (3) ranges over the single set $\{v\}$, giving $\sigma_0(v) = w(\text{cut}(\{v\}))$, the weighted degree, which separates items with distinct neighbourhoods. The identification count is therefore non-decreasing in the radius and strictly increasing wherever the overlap strictly increases. Section 12 reports both quantities. $\square$

Remark 3.7 (We predicted the opposite, and the sweep corrected us). Our stated expectation was that a coarse radius would identify more, by analogy with a matcher in which a wider neighbourhood discriminates more finely. The measurement is the reverse, and theorem 3.6 says why: in that setting the radius enlarges the *description* of each item, whereas here it enlarges the *domain of a minimisation* that different items come to share.

Enlarging a description separates; enlarging a shared minimisation merges. We record the correction, and we report the mean ball overlap beside the identification count, because a count without its mechanism would leave the direction unexplained and would invite the same mistake again.

Remark 3.8 (What theorem 3.5 rules out). A threshold can be tuned to a desired outcome, and its value is a free parameter a reader must take on trust. A resolution cannot be tuned in the same way, because its effect is monotone and its direction is a stated consequence rather than a fitted one. Section 12 reports the whole sweep rather than a chosen value, and a reader may read off the setting their question requires: a fine radius to tell items apart, a coarse one to group them.

Part II

Rungs and Ladders

4 Derived Power

Definition 4.1 (Rung and power). A *rung* carries exactly one datum, its *power* $\pi \in [0, 1]$. A rung of power 0 is *inert*; a rung of power 1 is *absolute*.

Remark 4.2 (A rung has no identity field). Theorem 4.1 is complete as stated: a rung is a number. It records nothing about what performs the step, how long it takes, or what passes through it. This is not brevity. A formalism in which rungs carried names would permit an observable that read the names, and the claim that identities do not enter would then be a claim about which observables one had chosen to write down rather than about the formalism. The implementation enforces the same discipline—the rung type has one field—so that a test of label-independence is not testing the labels.

Definition 4.3 (Derived power). Let Γ be a contact graph, v an item and ϱ a radius. The *derived power* of v at radius ϱ is

$$\pi_\varrho(v) = 1 - \frac{\beta_\varrho(v)}{\sigma_\varrho(v)}. \quad (4)$$

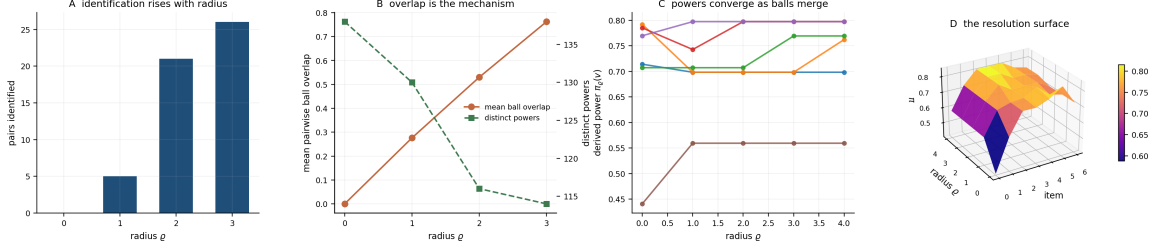


Figure 2: **Radius is a resolution parameter, and it runs in the direction opposite to the one we predicted.** (A) Pairs of items whose derived powers agree to 10^{-9} , against radius. Identification rises from zero at radius 0 to 26 at radius 3. Our stated prediction was that a coarse radius would identify more; the measurement is the reverse and theorem 3.7 records the correction. (B) The mechanism, which is why the direction is what it is. Mean pairwise ball overlap (solid) rises monotonically from 0.000 to 0.763, while the number of distinct powers (dashed, right axis) falls from 138 to 114. Balls of radius 0 are disjoint singletons and each item minimises over its own edges; balls of large radius coincide, so distinct items minimise over the same edge set and their powers converge. The overlap is plotted beside the count because a count without its mechanism would leave the direction unexplained. (C) Six individual power trajectories on one chain as the radius grows. Traces that begin apart come together and then flatten, which is the per-item form of the aggregate in (B). (D) The resolution surface: derived power over item and radius. The surface is ridged at small radius, where items differ, and flattens along the radius axis as the balls merge.

Lemma 4.4 (Derived power is a power). **4.1 Intensity**
 $\pi_\varrho(v) \in [0, 1)$ for every item and radius.

Proof. By theorem 3.4, $0 < \beta_\varrho(v) \leq \sigma_\varrho(v)$, so the ratio lies in $(0, 1]$ and $\pi_\varrho(v) \in [0, 1)$. The value 1 is not attained, which is the floor asserting itself: no single rung closes the whole gap, because closing it entirely would require separating v at zero cost. $\square$

Remark 4.5 (Reading eq. (4)). The quantity compares what it costs to separate v from the medium against the cheapest separation available anywhere in its neighbourhood. An item that is expensive to separate relative to its surroundings has high power: committing it closes much of the remaining distance. An item whose separation cost is already at the local floor has power near zero: there is nothing left to close. The floor appears in the numerator and not as a subtracted constant because what a rung closes is a *fraction* of the above-floor gap, and the floor is the part of the gap that cannot be closed.

Definition 4.6 (Intensive and extensive). Let f assign a real number to an item of a contact graph. Call f *intensive at radius ϱ* if $f(v)$ is unchanged by any modification of the graph outside $B_\varrho(v)$ that leaves the edges of $B_\varrho(v)$ and their medium edges intact, and *extensive* otherwise.

Theorem 4.7 (Derived power is intensive). π_ϱ is intensive at radius ϱ . Moreover both factors are required: the quantity $1 - \beta/\sigma_\varrho(v)$, in which the cut key is local but the normaliser is the global floor, is not intensive at any radius.

Proof. For the first claim, $\sigma_\varrho(v)$ minimises over subsets of $B_\varrho(v)$, and for such an S the cut $\text{cut}(S)$ consists of edges with exactly one endpoint in $S \subseteq B_\varrho(v)$. Every such edge is either internal to $B_\varrho(v)$, or joins $B_\varrho(v)$ to the medium, or joins $B_\varrho(v)$ to an item outside it; in the last case the outside endpoint is a neighbour of a ball member, hence lies in $B_{\varrho+1}(v)$, and the edge is determined by the ball's own incident edge set. A modification outside $B_\varrho(v)$ preserving those edges therefore preserves every term of the minimisation,

hence preserves $\sigma_\varrho(v)$. The same argument applies to $\beta_\varrho(v)$, which minimises over a subset of the same edges. The ratio is therefore unchanged.

For the second claim it suffices to exhibit a modification that changes β without touching $B_\varrho(v)$. Adjoin to Γ a fresh item u outside $B_\varrho(v)$, joined to the medium by an edge of weight $\beta/2$. The new graph is a contact graph, $B_\varrho(v)$ is untouched, so $\sigma_\varrho(v)$ is unchanged; but the global floor is now $\beta/2$, so $1 - \beta/\sigma_\varrho(v)$ strictly increases. Hence the quantity is not intensive. $\square$

Remark 4.8 (We got this wrong first, and the near-miss is the informative control). The construction we tried first was $1 - \beta/\sigma_\varrho(v)$: a local cut key normalised by the ambient floor, which is the floor the language already declares and therefore the one nearest to hand. It fails, and theorem 4.7 says why. We keep it as a control rather than deleting it, for a reason worth stating. A control that differs from the candidate in every respect shows only that *some* difference matters. This one differs in a single component and therefore localises the failure to the normaliser. In the measurements of section 12 the three quantities drift by exactly 0, by 1.6×10^{-2} and by 9.4×10^{-2} respectively, and the ordering of the middle term is the whole content of the diagnosis.

Corollary 4.9 (The floor is load-bearing only through positivity). *Every result in this section uses $\beta_\varrho(v)$ only through the facts that it is strictly positive and is attained on a finite set. Any other rule assigning a strictly positive local normaliser to each ball yields the same theorems with different constants. In particular no result determines the value β , and none is disturbed by rescaling every weight by a positive constant.*

Proof. Theorem 3.4 and theorem 4.4 use only positivity and attainment. Theorem 4.7 uses only that the normaliser is computed from the ball. Under $w \mapsto cw$ with $c > 0$ both β_ϱ and σ_ϱ scale by c and eq. (4) is unchanged. $\square$

5 Composition

Definition 5.1 (Ladder). A *ladder* is a finite sequence of rungs $\gamma_1, \dots, \gamma_n$ with power sequence $(\pi_1, \dots, \pi_n)$, applied in order, each to the state its predecessor produced.

Theorem 5.2 (Multiplicative composition). *Write r_i for the above-floor gap remaining after i rungs, with $r_0 > 0$ the initial gap. If each rung closes the fraction π_i of the gap it is presented with, then $r_i = (1 - \pi_i)r_{i-1}$, hence*

$$r_n = r_0 \prod_{i=1}^n (1 - \pi_i), \quad \pi(L) = 1 - \frac{r_n}{r_0} = 1 - \prod_{i=1}^n (1 - \pi_i). \quad (5)$$

Proof. The recurrence is theorem 4.1 rearranged. Induction on n gives the product. The composite power is the fraction of the initial gap closed, which is $1 - r_n/r_0$. $\square$

Corollary 5.3 (Bounded, monotone, and never complete). $\pi(L) \in [0, 1)$ whenever every $\pi_i < 1$; it is non-decreasing under appending a rung; and appending a rung of power 0 leaves it unchanged. Repeating a rung of power $\pi \in (0, 1)$ n times gives $1 - (1 - \pi)^n$, which tends to 1 and equals it at no finite n .

Proof. Each factor $(1 - \pi_i) \in (0, 1]$, so the product lies in $(0, 1]$ and the composite in $[0, 1)$. Appending multiplies the product by a factor at most 1, so the composite does not decrease; a factor of exactly 1 leaves it fixed. The repetition formula is eq. (5) with all powers equal, and $(1 - \pi)^n > 0$ for all finite n . $\square$

Proposition 5.4 (Sensitivity is transmission). $\frac{\partial \pi(L)}{\partial \pi_j} = \prod_{i \neq j} (1 - \pi_i) = \frac{P}{1 - \pi_j}$, where $P = \prod_i (1 - \pi_i)$ is the residual fraction. Within a ladder this is strictly increasing in π_j : control lies at the highest-power rung.

Proof. Differentiate eq. (5): only the j -th factor depends on π_j and it does so with derivative -1 , so the derivative of the product is $-\prod_{i \neq j} (1 - \pi_i)$ and the composite's derivative is its negation. Dividing and multiplying by $(1 - \pi_j)$ gives the second form, which is increasing in π_j at fixed P . $\square$

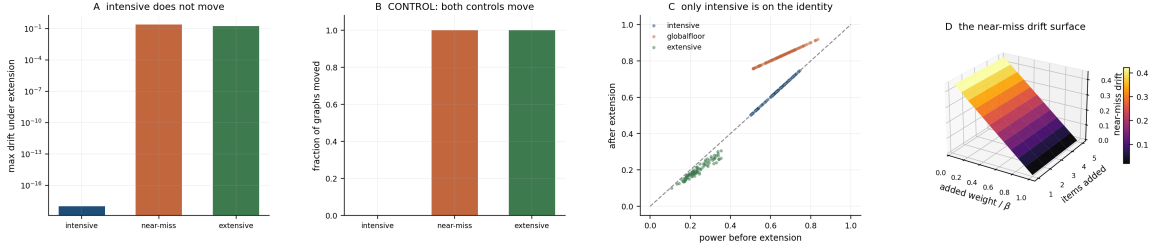


Figure 3: **Intensity requires a local normaliser, and the near-miss control is what shows it.** (A) Maximum drift in derived power when the graph is extended far from the item measured, over 120 graphs, on a logarithmic axis. The intensive candidate does not move at all—the bar is at the floor of the axis because the value is exactly zero, not merely small. Both controls move by more than 10^{-1} . (B) The control, and the reason panel A carries information. Fraction of graphs on which each candidate moved: 0/120 for the intensive candidate and 120/120 for both controls. Had no control moved, the test would have been measuring something other than intensity and would be reported as non-discriminating. (C) Power before against power after extension, for all three candidates. The intensive points lie on the identity exactly; the near-miss lies visibly above it, because the global floor fell and every power rose with it; the extensive points scatter below. The near-miss differs from the candidate in one component only—the normaliser—so its displacement localises the failure rather than merely exhibiting one. (D) Near-miss drift over the weight of the added edge and the number of items added. Drift is governed by the added *weight* and is nearly flat in the *count*, which is the signature of the global floor: one small distant edge is enough, and further additions change little.

Remark 5.5 (The direction is the opposite of the naive expectation). The expectation is that the weakest rung is the bottleneck and therefore the place where improvement pays. Theorem 5.4 says the opposite, and the reason is that the quantity being maximised is *transmission* rather than slack: a marginal gain at rung j reaches the output only through the remaining rungs, and the product $\prod_{i \neq j} (1 - \pi_i)$ is what survives that passage. Improving a weak rung buys a large local change that the rest of the ladder then discards. This is a prediction and it is sharp enough to fail; section 12 reports it measured.

5.1 Order dependence, stated as measured

Equation (5) is symmetric in its arguments, so permuting a *fixed* power sequence cannot change the composite. That fact is a property of the formula and carries no information about the derivation, and a test built on it cannot fail. We say so explicitly because such a test is easy to write and easy to mistake for

evidence.

Remark 5.6 (A test that cannot fail is not evidence). Permuting a fixed multiset of powers and observing that the composite is unchanged verifies the commutativity of multiplication. Every candidate definition of power passes it, including deliberately extensive ones. Our suite runs this comparison and reports it as NON-DISCRIMINATING, excluded from the score, with this reason recorded (section 12). It is retained because a reader cannot distinguish a passing control from a vacuous one by inspecting a table of successes.

The test that can fail derives the powers *sequentially*: each rung’s power is computed on the graph as its predecessors left it, so a different order presents each rung with a different graph.

Definition 5.7 (Sequential derivation). Given an ordering $v_1, \dots, v_n$ of items, the *sequentially derived* power sequence is obtained by setting $\pi_i = \pi_{\varrho}(v_i)$ evaluated in Γ_{i-1} , where $\Gamma_0 = \Gamma$ and Γ_i is Γ_{i-1} with the weakest item–item contact incident to v_i removed.

Proposition 5.8 (Order dependence is graded, not abolished). *Under sequential derivation the composite power is in general order-dependent, for intensive and extensive candidates alike. Intensity reduces the dependence but does not remove it, and the reduction is monotone in locality.*

Proof and measurement. That some dependence survives is immediate: theorem 5.7 modifies the graph between derivations, and theorem 4.7 guarantees invariance only against modifications *outside* the ball. A removal incident to v_i lies inside the balls of v_i 's neighbours, so it is exactly the case the theorem does not cover.

The size of the residual dependence is a measurement, not a theorem. Over all 720 orderings of a six-item chain, the number of distinct composite values was 47 for the intensive candidate, 118 for the near-miss and 421 for the extensive control, with spreads 1.07×10^{-2} , 1.09×10^{-2} and 1.04×10^{-1} . At radius 0 the intensive count falls to 11. The ordering of these numbers is the content of the proposition; their values are properties of the instance and we do not generalise them. $\square$

Remark 5.9 (Why we state the graded result). We set out to prove that intensity buys exact order-independence. It does not, and the measurement above is what showed it. Reporting the clean claim would have required either restricting to ladders whose balls never overlap—a condition satisfied by few interesting processes—or declining to mutate the graph, which would have made the test unable to fail. The graded statement is weaker and true, and it comes with a mechanism: dependence enters exactly where commitment reaches inside a neighbouring ball.

6 Two Composition Rules, Neither Determining the Other

Residue chaining and ladder composition are both defined on a chain of cuts. This section shows they are independent, which is what justifies keeping both and what makes

theorem 1.1 an amendment rather than a re-statement.

Definition 6.1 (The two aggregates). For a chain $C = (S_1, \dots, S_n)$ with residues $\rho_i = w(\text{cut}(S_i))$ and powers π_i , the *extensive aggregate* is $R(C) = \sum_i \rho_i$ and the *intensive aggregate* is $\pi(C) = 1 - \prod_i (1 - \pi_i)$.

Proposition 6.2 (Independence). *Neither aggregate determines the other. There exist chains with equal R and different π , and chains with equal π and different R .*

Proof. For the first, take two chains of two cuts each, with residues $(2\beta, 2\beta)$ in both cases, so $R = 4\beta$ for each. Let the first close gap fractions $(0.5, 0.5)$ and the second $(0.9, 0.1)$; the composites are 0.75 and 0.91. Residue is the boundary committed and power is the distance closed, and nothing ties them: a cut may commit much boundary while closing little gap.

For the second, take chains with powers $(0.5, 0.5)$ and (0.75) , both of composite 0.75, whose residues are 2β and β respectively; the extensive aggregates differ because the chains have different lengths. $\square$

Corollary 6.3 (The reversal lemma does not transfer). *The reversal lemma establishes that R is invariant under reversing a chain. The same holds for π by symmetry of eq. (5). Consequently the ladder inherits the limitation the medium paper draws from that lemma: no predicate definable from a ladder's powers alone orients a process, and orientation must still be sought in the medium.*

Proof. Symmetry of the product under permutation gives the invariance. The negative consequence is then the same argument as for R : a quantity invariant under reversal cannot distinguish a chain from its reversal. $\square$

Remark 6.4 (We record the limitation rather than leaving it to be found). Theorem 6.3 is a limitation of the construct introduced in this paper, and a reader who noticed the symmetry of eq. (5) would reach it unaided. The ladder answers “how much of the distance did this step close” and does not answer “which way does the process run”; the second question

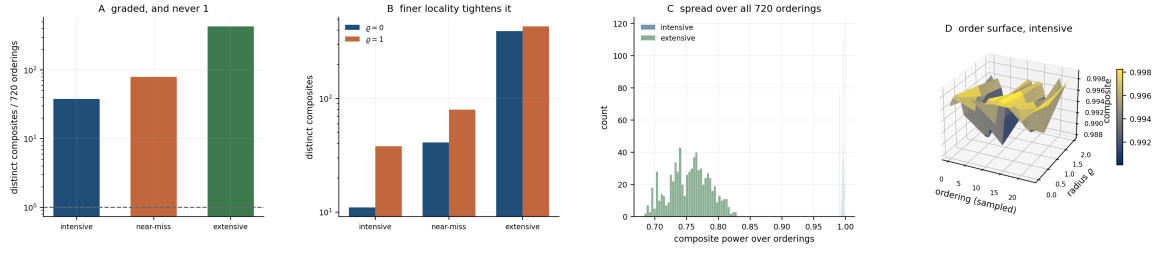


Figure 4: **Order dependence under sequential derivation is graded, not abolished.** (A) Distinct composite powers over all 720 orderings of a six-item chain at radius 1, on a logarithmic axis, with the line at 1 marking exact order independence. The counts are 38, 80 and 432; intensivity reduces dependence by an order of magnitude relative to the extensive control and no candidate attains 1. The dashed line is drawn because the claim we set out to prove was that the intensive bar would sit on it. (B) The same counts at radius 0 and radius 1. Finer locality tightens the intensive count from 38 to 11 without reaching 1: dependence survives because commitment mutates edges inside the balls of neighbouring items, which is exactly the case the intensivity theorem does not cover. (C) The distributions behind the counts: composite power over all 720 orderings for the intensive candidate and the extensive control. The intensive distribution is narrow and the extensive one broad, and neither is a single spike. (D) Composite power over sampled orderings and radius for the intensive candidate. The surface is corrugated along the ordering axis at every radius—that corrugation is the residual order dependence—and its amplitude falls as the radius falls.

was answered elsewhere and the ladder does not reopen it.

Part III

The Language

7 Syntax

We add four productions. The remainder of the grammar is unchanged, and we reproduce only what the additions touch.

Definition 7.1 (Added grammar).

```
ladderExpr ::= ladder [ runList ] [ at
    radius number ]
runList ::= runListItem { , runListItem }
runListItem ::= power number | rung ident
    | derived ident
climbExpr ::= climb expr with
    ladderExpr [ reach number ]
observeLad ::= observe expr as power
```

The keyword set gains `ladder`, `rung`, `power`, `derived`, `radius`, `climb` and `reach`. The non-terminal *expr* and all existing productions are untouched.

Remark 7.2 (Filling a gap rather than widening one). The host specification declares a `medium` block by example and gives it no production, and its type grammar omits `Role` although a rule concludes at that type. We do not repair those omissions here—they belong to the papers that introduced them—but we avoid adding to them: every construct below has a production in theorem 7.1, a type in theorem 8.1, a typing rule in section 8 and a reduction rule in section 9.

Example 7.3 (A ladder over a folded receiver).

```
receiver bio
floor 3.7e-4

medium cytosol {
    ambient "H2O" = 55.5
    exchange 1.0e-3
}

P := fold "1CRN"

-- powers declared explicitly
L1 := ladder [ power 0.45, power 0.30,
    power 0.55, power 0.20 ]

-- powers derived from the receiver at
    a declared resolution
```

```

L2 := ladder [ derived P ] at radius 1

observe L1 as power -- free: reads
    committed boundary
climb P with L1 reach 0.80 -- commits;
    M advances by four

```

The first `observe` is free. The `climb` is not: it performs four commitments and advances the clock by four.

8 Types

Definition 8.1 (Added types). The type grammar gains two constructors:

$$\tau ::= \dots \mid \text{Rung} \mid \text{Ladder}.$$

A value of type `Rung` carries a power in $[0, 1]$; a value of type `Ladder` carries a finite sequence of such. Both are values in the sense of the host judgement and carry a floor annotation.

The judgement is the host's, $\Gamma \vdash e : \tau @ \beta$, read: in context Γ fixing the ambient floor and receiver, e has type τ and is individuated at floor $\beta > 0$.

Rule 8.2 (Rung introduction).

$$\frac{\Gamma \vdash p : \text{Scalar} @ \beta \quad 0 \leq p \leq 1}{\Gamma \vdash \text{power } p : \text{Rung} @ \beta}$$

A power outside $[0, 1]$ is ill-typed. The bound is not a runtime check: by theorem 4.4 a derived power satisfies it by construction, and a declared one is checked where it is written.

Rule 8.3 (Derived rung).

$$\frac{\Gamma \vdash v : \text{Leaf}_\kappa @ \beta \quad \varrho \in \mathbb{N} \quad \mathbf{m} \text{ declared}}{\Gamma \vdash \text{derived } v \text{ at radius } \varrho : \text{Rung} @ \beta}$$

The premise $\mathbf{m}$ declared is the same side condition the host requires for role observation, and for the same reason: eq. (2) measures against the medium, so without a medium the power is undefined rather than zero.

Rule 8.4 (Ladder formation).

$$\frac{\Gamma \vdash \gamma_i : \text{Rung} @ \beta \quad (1 \leq i \leq n)}{\Gamma \vdash \text{ladder } [\gamma_1, \dots, \gamma_n] : \text{Ladder} @ \beta}$$

A ladder is well-typed at the floor its rungs share. Rungs derived at different radii may be combined; the radius is a property of a derivation, not of a rung.

Rule 8.5 (Power observation).

$$\frac{\Gamma \vdash L : \text{Ladder} @ \beta}{\Gamma \vdash \text{observe } L \text{ as power} : \text{Scalar} @ \beta}$$

Observing a composite power commits no new cut: the computation reads powers already carried by rungs already formed, so the clock M does not advance. This is free for exactly the reason role observation is free—it individuates nothing new—and theorem 9.5 states that reason as a criterion.

9 Operational Semantics

Configurations are those of the host: $\langle \Gamma_G, M, e \rangle$, the contact graph, the committed-cut count, and the expression under evaluation.

Rule 9.1 (E-Power: forming a rung is free).

$$\frac{0 \leq p \leq 1}{\langle \Gamma_G, M, \text{power } p \rangle \rightarrow \langle \Gamma_G, M, v_{\text{Rung}}(p) \rangle}$$

No boundary is committed and M does not advance: declaring a number individuates nothing.

Rule 9.2 (E-Derive: deriving a rung is free).

$$\frac{\pi_\varrho(v) = 1 - \beta_\varrho(v)/\sigma_\varrho(v)}{\langle \Gamma_G, M, \text{derived } v \text{ at radius } \varrho \rangle \rightarrow \langle \Gamma_G, M, v_{\text{Rung}}(\pi_\varrho(v)) \rangle}$$

The derivation reads edge weights already present in Γ_G and computes a minimum over them. It commits no cut, so M does not advance. The graph is read and not written.

Rule 9.3 (E-Observe-Power: reading a composite is free).

$$\frac{L = [\gamma_1, \dots, \gamma_n]}{\langle \Gamma_G, M, \text{observe } L \text{ as power} \rangle \rightarrow \langle \Gamma_G, M, v_{\text{Scalar}}(1 - \prod_i (1 - \pi_i)) \rangle}$$

Rule 9.4 (E-Climb: climbing commits, once per rung).

$$\frac{L = [\gamma_1, \dots, \gamma_n] \quad S_i \text{ admissible} \quad w(\text{cut}(S_i)) \geq \beta \quad r_n = r_0 \prod_i (1 - \pi_i)}{\langle \Gamma_G, M, \text{climb } x \text{ with } L \rangle \rightarrow \langle \Gamma_G \oplus \bigcup_i \text{cut}(S_i), M + n, v_{\text{Result}} \rangle}$$

Climbing performs one commitment per rung: the clock advances by n , and each commitment deposits residue at least the floor. This is the sole rule of the extension that is not free.

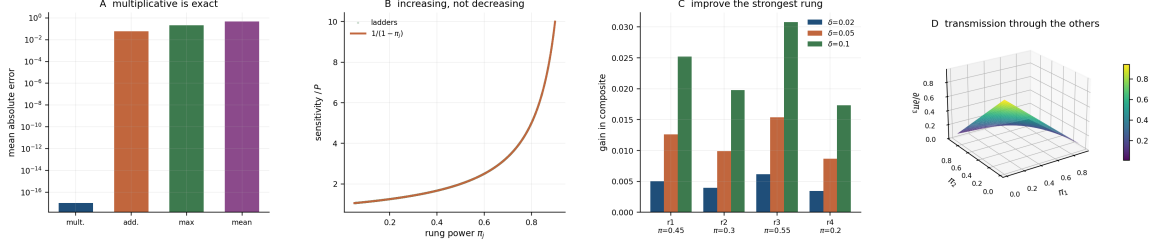


Figure 5: **Composition is multiplicative, and control lies at the strongest rung.** (A) Mean absolute error of four candidate composition laws against a rung-by-rung simulation, over 4000 ladders, on a logarithmic axis. The multiplicative law is exact to machine precision; the additive, maximum and mean laws err by 5.9×10^{-2} , 2.1×10^{-1} and 4.6×10^{-1} . All four are scored: scoring only the favoured law would not have distinguished it from the others. (B) Sensitivity normalised by the ladder’s residual fraction, against rung power, over 700 random ladders. Every point lies on $1/(1 - \pi_j)$, which is exact, since sensitivity is $P/(1 - \pi_j)$. The normalisation is necessary because raw sensitivity also depends on the other rungs, which obscures the relationship across ladders of differing composition. (C) Gain in composite power from a fixed improvement applied to each rung in turn, at three improvement sizes, for the four-rung ladder (0.45, 0.30, 0.55, 0.20). Rung 3, of highest power, returns most at every size and rung 4, the weakest, returns least. Improving the bottleneck is the worse investment, by roughly a factor of two at $\delta = 0.1$. (D) Sensitivity to the third rung of a three-rung ladder, over the powers of the other two. The surface is $(1 - \pi_1)(1 - \pi_2)$: what a marginal gain buys is set by transmission through the remaining rungs, which is why the quantity being maximised is transmission and not slack.

Theorem 9.5 (A criterion for freeness). *A reduction rule may leave M unchanged if and only if it commits no cut, that is, if it neither adds edges to the committed boundary nor requires a boundary not already committed. Under this criterion theorem 9.1, theorem 9.2 and theorem 9.3 are free and theorem 9.4 is not, and the role observation of the host is free.*

Proof. ($\Leftarrow$) If a rule commits no cut then the committed boundary and the cut count are both unchanged, by the definition of the counter as a count of cut events. Leaving M fixed is therefore consistent with clock monotonicity, which forbids only *decrease* and requires increase only *at a cut event*.

($\Rightarrow$) If a rule committed a cut and left M unchanged, two configurations with the same graph and the same counter would differ in the boundary committed, contradicting that the counter records cut events.

The classification is then immediate. Theorem 9.1 constructs a value from a literal; theorem 9.2 computes minima over existing

weights; theorem 9.3 multiplies numbers carried by values already formed. None writes to Γ_G . Theorem 9.4 writes $\Gamma_G \oplus \bigcup_i \text{cut}(S_i)$ and advances by n . $\square$

Remark 9.6 (The criterion is the argument of the host, generalised). The host licenses one free observation with a one-sentence reason: the comparison reads boundary that is already committed, so it individuates nothing new. Theorem 9.5 does not weaken that reason, it promotes it. What was a licence granted to one rule becomes a criterion that any rule may be tested against. This is the sense in which theorem 1.2 is a strengthening rather than a concession: a uniqueness claim is replaced by a decidable test, and role observation passes it.

Theorem 9.7 (Progress). *If $\Gamma \vdash e : \tau @ \beta$ and e is not a value, some reduction applies.*

Proof. By cases on the added forms. **power** p is well-typed only with $0 \leq p \leq 1$, which is the premise of theorem 9.1. **derived** v is well-typed only with $\mathbf{m}$ declared, and then

$\sigma_\varrho(v)$ is a minimum over a finite non-empty family, hence defined, and by theorem 3.4 strictly positive, so theorem 9.2 applies. A ladder of well-typed rungs is a value. The form `observe  $L$  as power` applies theorem 9.3 unconditionally. For `climb`, admissibility of each S_i follows from the typing of the rungs, and the residue side condition holds by theorem 2.3, so theorem 9.4 applies. Existing forms are covered by the cases of the host, which the extension does not modify. $\square$

Theorem 9.8 (Preservation). *If $\Gamma \vdash e : \tau @ \beta$ and $\langle \Gamma_G, M, e \rangle \rightarrow \langle \Gamma'_G, M', v \rangle$, then v has type τ and floor β , with β unchanged and $M' \geq M$.*

Proof. Each added rule produces a value of the type its typing rule assigns: theorem 9.1 and theorem 9.2 produce `Rung`, theorem 9.3 a `Scalar`, theorem 9.4 a `Result`. The floor is a component of the context and no added rule modifies it. Three rules leave M fixed and one increments it by $n \geq 0$, so $M' \geq M$. Theorem 9.4 mutates Γ_G ; the mutation retains the medium and every weight remains strictly positive, so theorem 2.3 continues to apply to the mutated graph and the floor annotation remains valid. $\square$

Corollary 9.9 (Soundness). *A well-typed program in the extended language does not get stuck, and every commitment it performs deposits residue at least $\beta > 0$.*

Theorem 9.10 (Clock monotonicity is preserved). *M is non-decreasing along every evaluation of the extended language, and increases strictly at every cut event. Re-evaluating a `climb` is a new sequence of commitments at strictly higher M , never a cached retrieval; re-evaluating a free form is not.*

Proof. No added rule decrements M . Theorem 9.4 increments by the number of cut events it performs, and the three free rules perform none, so by theorem 9.5 their leaving M fixed is consistent with the requirement that increase occur at every cut event. A cache returning a previous `climb` result without incrementing would contradict theorem 9.4, which increments unconditionally. $\square$

Remark 9.11 (A repair to a proof of the host, not to its theorem). The proof of clock monotonicity in the host asserts in passing that every value-producing rule increments M . That parenthetical is already false in the host, since role observation produces a value of type `Role` without incrementing, and it is false three more times here. The theorem itself is unaffected: what it requires is that no rule decrements and that every *cut event* increments. We state the repaired form, and observe that theorem 9.5 supplies exactly the side condition the original proof left implicit.

Proposition 9.12 (The extension is conservative on existing programs). *Let p be a program containing none of the added keywords. Then p has the same typing derivations and the same reductions in the extended language as in the host, and yields the same value at the same final M .*

Proof. The added typing rules have conclusions whose subjects are `power`, `derived`, `ladder`, `climb`, or `observe ... as power`; the added reduction rules likewise. None of these forms occurs in p , so no added rule is applicable at any point in a derivation or reduction of p . No existing rule is modified. Hence the derivations and reduction sequences are literally the same objects, and the final value and counter agree. $\square$

10 Static Analysis and the Second Refusal

10.1 Deciding reachability before execution

Proposition 10.1 (Reachability is decidable statically). *Whether a ladder with declared powers $(\pi_1, \dots, \pi_n)$ attains a declared target π^* is decidable in $O(n)$ arithmetic operations without evaluating any `climb`, and therefore without committing any cut or advancing M .*

Proof. Compute $\prod_i (1 - \pi_i)$ in n multiplications and compare 1 minus it against π^* . By theorem 5.2 this is exactly the value a `climb` would reach. The computation reads declared numbers and touches no graph. $\square$

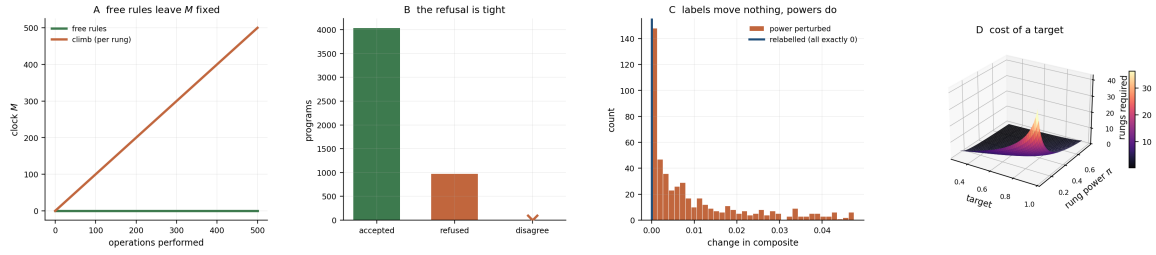


Figure 6: **The clock, the refusal, and label independence with its control.** **(A)** The clock M under the two regimes. Free rules—forming a rung, deriving a rung, reading a composite—leave M flat at zero however many are performed; a climb advances it once per rung. This is the operational content of the freeness criterion, which licenses a free rule exactly when it commits no cut. **(B)** The static verdict against execution on 5000 generated programs: 4032 accepted, 968 refused, and zero disagreements, marked by the cross at zero. Both verdicts occur in quantity, so the test could have failed in either direction; a suite in which every program was accepted would not have tested rejection. **(C)** Label independence and the control that gives it content. The vertical line at zero is the distribution of changes under relabelling: every one is exactly zero, since a rung has no identity field and the relabelling acts on annotations external to the formalism. The histogram is the control, in which a rung *power* is perturbed instead; those changes are entirely positive and well away from zero. Without the control the first result would be consistent with a constant observable and would carry no information. **(D)** Rungs required to reach a target, over target and rung power. The surface diverges as the target approaches unity and as the rung power approaches zero, and is low only where strong rungs meet modest targets. This is the cost model behind the refusal: what a target costs is set jointly by how demanding it is and by how much each step accomplishes.

Corollary 10.2 (Cost of a target). *Reaching $\pi^* < 1$ with copies of a rung of power $\pi \in (0, 1)$ requires exactly $n = \lceil \log(1 - \pi^*) / \log(1 - \pi) \rceil$ rungs, and no finite number suffices for $\pi^* = 1$.*

Proof. Solve $1 - (1 - \pi)^n \geq \pi^*$ for n ; both logarithms are negative so the inequality direction is preserved on dividing. Minimality is the ceiling. For $\pi^* = 1$ we would need $(1 - \pi)^n \leq 0$, impossible for finite n . $\square$

10.2 The refusal

The host holds that a representation earns a positive result only if some presentable object would have received a negative one, and that a framework with no refusals has an empty defined class. The medium paper discharged this with two refusals, to individuate and to orient. The ladder supplies a third, which is the second refusal of a *construct* rather than of an observation.

Definition 10.3 (Refusal to climb). A ladder L with declared target π^* for which $1 - \prod_i (1 - \pi_i) < \pi^*$ is *refused*: the program does not evaluate, the verdict is *subfloor* naming the shortfall $\pi^* - \pi(L)$, and no cut is committed. A ladder with no rungs is refused with verdict *empty*.

Remark 10.4 (The verdict name is taken, not coined). *subfloor* is an existing verdict label in the reference implementation of the surrounding framework, where it already means that a quantity fell below a declared floor. We reuse it rather than introduce a synonym. Similarly we do not reuse the label *inert*, which is bound in that implementation to a different meaning, and we say *label-independent* where an unqualified reading of “inert” might collide.

Theorem 10.5 (The refusal is tight). *No ladder refused by theorem 10.3 could have reached its declared target. The refusal therefore excludes no program that would have succeeded, and every refused program would have failed at run time.*

Proof. Suppose L is refused, so $1 - \prod_i (1 - \pi_i) < \pi^*$. By theorem 5.2 the gap after all n rungs is $r_0 \prod_i (1 - \pi_i)$, so the fraction of

the initial gap closed by executing L in full is exactly $1 - \prod_i (1 - \pi_i)$, which is below π^* by hypothesis. Executing the ladder therefore terminates short of the target. Conversely a ladder not refused has composite at least π^* and reaches it. The static verdict and the executed outcome therefore agree on every ladder, which is the content of tightness. $\square$

Theorem 10.6 (The refusal is non-vacuous). *Both refusals of theorem 10.3 are attainable by ladders the language can construct.*

(i) *The ladder `ladder[power 0.1, power 0.1]` with target 0.9 has composite $0.19 < 0.9$ and is refused with *subfloor*.*

(ii) *The ladder `ladder[ ]` is refused with *empty*.*

Moreover refusal is not universal: the ladder `ladder[power 0.5, power 0.6]` with target 0.7 has composite $0.8 \geq 0.7$ and is accepted. Hence the predicate separates presentable ladders in both directions.

Proof. Direct computation of eq. (5) in each case: $1 - (0.9)(0.9) = 0.19$; the empty product is 1 so the composite is 0; and $1 - (0.5)(0.4) = 0.8$. Each ladder is well-typed by theorem 8.2 and theorem 8.4, hence presentable. $\square$

Remark 10.7 (Why the third clause is part of the theorem). A refusal that fired on every input would classify nothing, exactly as a refusal that never fired classifies nothing. Theorem 10.6 therefore exhibits an accepted ladder alongside the refused ones. The corresponding requirement on the validation suite is that both verdicts occur in quantity, and section 12 reports the counts rather than only the agreement rate.

Proposition 10.8 (Saturation diagnostic). *If every rung of a ladder of length n has power at most $\pi_{\max}$, and $1 - (1 - \pi_{\max})^n < \pi^*$, then the ladder is refused whatever its actual powers. The diagnostic is sound and is computed from n and $\pi_{\max}$ alone.*

Proof. The composite is monotone non-decreasing in each power by theorem 5.3, so it is maximised at $\pi_i = \pi_{\max}$ for all i , giving

$1 - (1 - \pi_{\max})^n$. If even that is below the target, no assignment of powers within the bound reaches it. $\square$

Part IV

Measurement

11 Protocol

The suite is three modules and a runner, at `validation/` beside this paper. Every number quoted below is read from the JSON the runner writes; none is entered by hand.

Scoring policy. A check contributes to the score only if it could have failed. A check whose own construction shows the statistic cannot discriminate is marked NON-DISCRIMINATING, excluded from the score, and its reason recorded. A claim the suite cannot bear on is marked NOT TESTED and excluded. Both categories are reported rather than omitted, because a reader cannot distinguish a passing control from a vacuous one by inspecting a table of successes.

Negative controls. Four checks are named with the prefix NEGATIVE CONTROL so that they can be harvested by name, following the convention of the host suite. Each pairs with a positive claim and would falsify it if it behaved differently.

Totals. Twenty-three checks run. Twenty-one are scored and all twenty-one pass; one is non-discriminating; one is not tested. Four of the scored checks are negative controls.

12 Results

12.1 The substrate

The floor is computed rather than assumed on four instances, including an adversarial one whose smallest weight is 10^{-12} ; every instance returns a strictly positive floor. The failure branch is exhibited separately: the sequence $1/n$ clears every ε down to 10^{-9} , so its infimum

is zero, and truncating the same sequence to ten stages restores a floor at 0.1. This is the content of theorem 2.4 made visible rather than left tacit.

The local separation cost of eq. (3) is checked against exhaustive enumeration of every bipartition on forty graphs: at a radius covering the graph the two agree to within 10^{-12} , which is a check of the implementation against ground truth rather than a restatement of the definition.

12.2 The radius sweep, and the direction we predicted wrongly

Table 1: The radius sweep. Identification rises with the radius, and the mean pairwise ball overlap rises with it too. The overlap is the mechanism: theorem 3.6.

Radius ϱ	0	1	2	3
Pairs identified	0	5	21	26
Mean ball overlap	0.000	0.276	0.530	0.763
Distinct powers	138	130	116	114

Table 1 is the measurement behind theorem 3.6. Three things are worth reading off it. First the radius is not inert: the counts move. Second the direction is the opposite of what we predicted, and theorem 3.7 records that we predicted wrongly. Third the overlap column is reported beside the identification column because it is the mechanism; a count without its mechanism would leave the direction unexplained and would invite the same mistake again.

12.3 Intensity, and the near-miss

Table 2 is the measurement behind theorem 4.7. The intensive candidate is invariant *exactly*—not to within a tolerance, but at zero drift on every one of the 120 graphs—which is what an invariance proof predicts and what a numerical coincidence would not deliver. Both controls move on every graph, so the test discriminates; had neither moved it would have been measuring something other than intensity and we would have reported it as non-discriminating.

Table 2: Drift of three candidate powers when the graph is extended far from the item measured, over 120 graphs. The intensive candidate does not move at all; both controls move on every graph.

Candidate	Max drift
$1 - \beta_\varrho(v)/\sigma_\varrho(v)$ (intensive)	0
$1 - \beta/\sigma_\varrho(v)$ (near-miss)	2.50×10^{-1}
$\sigma_\varrho(v)/\Omega$ (extensive)	1.78×10^{-1}

The near-miss is the informative control. It differs from the candidate in one component—the normaliser—and its drift is the largest of the three. That localises the failure: the cut key was already local, and locality of the cut key alone buys nothing.

12.4 Order dependence

Table 3: Distinct composite powers over all 720 orderings of a six-item chain under sequential derivation (theorem 5.7), with the spread of those composites.

Candidate	Radius 0		Distinct
	Distinct	Spread	
intensive	11	3.67×10^{-3}	38
near-miss	41	2.00×10^{-3}	80
extensive	392	9.22×10^{-2}	432

Table 3 is the measurement behind theorem 5.8. The ordering of the distinct counts is the result: $38 < 80 < 432$ at radius 1, and $11 < 41 < 392$ at radius 0, so intensivity reduces order dependence by an order of magnitude relative to the extensive control, and finer locality reduces it further. Exact independence would be a count of 1 and is not attained by any candidate.

One column runs against the pattern and we report it rather than omit it: the *spread* of the near-miss is smaller than that of the intensive candidate at both radii, although its distinct count is twice as large. The two statistics measure different things—how many values occur, and how far apart the extremes lie—and only the count separates the candi-

dates consistently. We therefore rest theorem 5.8 on the count and record the spread beside it, rather than quoting whichever statistic favours the conclusion.

12.5 The language

Table 4: Composite powers against three alternatives, over 4000 generated ladders, each scored against a rung-by-rung simulation.

Law	Mean absolute error
$1 - \prod_i (1 - \pi_i)$ (multiplicative)	0
$\min(1, \sum_i \pi_i)$ (additive)	5.94×10^{-2}
$\max_i \pi_i$	2.07×10^{-1}
$\frac{1}{n} \sum_i \pi_i$ (mean)	4.57×10^{-1}

All four laws are scored. Scoring only the favoured one would not have distinguished it from the others, and the comparison is the test.

Sensitivity is checked two ways over 3000 ladders: the closed form $P/(1 - \pi_j)$ matches the derivative on every ladder, and a finite-difference experiment that does not use the closed form identifies the same rung as returning most for a fixed improvement on every ladder. The second check is independent of the first, so the counter-intuitive direction of theorem 5.5 is established without assuming the formula that predicts it.

Freeness and the clock behave as theorem 9.5 requires: five thousand free operations leave M at zero, a climb of three rungs advances it to exactly three and records three residues each at least the floor, and five hundred identical commitments increment M by exactly one apiece with no caching. The paired negative control confirms that a commitment does register, so the first result is about freeness and not about a counter that never moves.

12.6 The refusal

Over 5000 generated programs the static verdict and the executed outcome agree in every case, with 4032 accepted and 968 rejected. Both verdicts occur in quantity, so the test could have failed in either direction; a suite in

which every program was accepted would not have tested rejection. This is theorem 10.5 measured.

Three negative controls attend the refusal. It fires on a presentable ladder that cannot reach its target and on an empty ladder, and declines to fire on a ladder that can—so the predicate separates in both directions. A refused ladder commits nothing: no residue is recorded and the graph is unchanged, which is the operational content of a refusal that declines to individuate. And the verdict type rejects a payload that does not match its label, so a refusal cannot smuggle out a result.

12.7 What the suite does not establish

One check is recorded NOT TESTED: whether a derived power corresponds to any rate measured in a laboratory. The powers here are derived from a contact graph, which is itself a model, and no check in this suite bears on the empirical question. We record it rather than leave the omission to be noticed.

One check is recorded NON-DISCRIMINATING: permuting a fixed multiset of powers. Every candidate returns exactly one distinct composite, because the composition law is symmetric in its arguments. The check cannot fail and is excluded from the score, with that reason stored in the results file.

13 Falsifiable Predictions

Each prediction is stated with the observation that would refute it.

P1. Multiplicative composition. Composite power follows eq. (5) and not an additive, maximum or averaging law. *Failure mode:* measured composite tracking $\sum_i \pi_i$ or $\max_i \pi_i$ across synthesised ladders.

P2. Intensity requires a local normaliser. A power built from a local cut key and the global floor is not intensive. *Failure mode:* the near-miss

candidate showing zero drift under extension, which would mean the global floor is not doing the damage we attribute to it.

P3. Order dependence is graded. Under sequential derivation the intensive candidate admits strictly fewer distinct composites than the extensive one, and strictly more than one. *Failure mode:* either exactly one—which would make the graded claim too weak—or no separation between candidates, which would make it empty.

P4. Radius is a resolution. Identification is monotone in the radius and tracks ball overlap. *Failure mode:* a flat sweep, showing the parameter is inert; or identification moving opposite to overlap, showing the mechanism is not the one we give.

P5. Control at the strongest rung. Sensitivity is $P/(1 - \pi_j)$, maximised at the highest-power rung. *Failure mode:* a finite-difference experiment ranking the weakest rung first. This prediction is deliberately counter-intuitive and is the sharpest in the set.

P6. Freeness. A rule leaves M unchanged exactly when it commits no cut. *Failure mode:* a rule that commits a cut while leaving M fixed, or a free rule that advances it.

P7. Tightness. The static refusal agrees with execution on every ladder, and both verdicts occur. *Failure mode:* any disagreement, or a suite in which nothing is refused.

14 Relation to Existing Accounts

Metabolic control analysis. Control coefficients summing to unity are the closest existing account of where control sits in a chain. The sensitivity of theorem 5.4 is not a control coefficient: it is a derivative of a composite

fraction rather than of a flux, and it is not normalised to sum to one. The two therefore need not agree, and theorem 5.5 is a prediction that can be tested against a measured control distribution rather than a restatement of one.

Rate-limiting steps. The notion that one step limits a process is recovered here as the rung of highest power, and theorem 5.4 says that this is also where improvement pays. This inverts the usual advice to improve the slowest step, and the inversion is a consequence of measuring fractions of a gap rather than rates.

Reliability and series systems. The law $1 - \prod(1 - \pi_i)$ is formally the probability that at least one of n independent events occurs, and appears in reliability theory. We do not claim the powers are probabilities; they are fractions of a gap, and the law is derived in theorem 5.2 from the recurrence rather than assumed from an independence hypothesis.

Provenance monoids. The composition of provenance tags under max over an ordered three-element set is an idempotent commutative monoid with an absorbing element. Ladder composition is the same shape without idempotence: identity 0, absorbing element 1, commutative, associative. The parallel is worth stating because the accompanying results—composition is a fold, a chain cannot regress, and the aggregate inherits monotonicity only under a stated hypothesis—transfer directly, and because it locates what is distinctive here as the loss of idempotence rather than the algebra.

Screens versus identity tests. A construct that assigns each item an invariant, and identifies items whose invariants agree, is a screen and not an identity test: agreement is necessary for sameness and not sufficient. Theorem 14.1 states this for ladders explicitly.

Proposition 14.1 (A ladder is a screen, not an identity test). *Equality of composite power does not entail equality of power sequences.*

Ladders with different power sequences may share a composite.

Proof. (0.5, 0.5) and (0.75) both give composite 0.75 by eq. (5). The full power sequences differ, and the intermediate gaps differ, so an observer recording the whole trajectory distinguishes them; an observer recording only the composite does not. $\square$

Remark 14.2 (Why the limitation is stated rather than buried). Theorem 14.1 is the natural objection to any claim that a ladder characterises a process, and it is correct. An experiment resolving only end-to-end efficiency underdetermines the ladder. What the ladder determines is what an observer of the whole trajectory sees, and section 12 reports the trajectory rather than the summary wherever both are available.

15 Limitations

1. Everything downstream of theorem 2.3 requires the contact graph to be finite (theorem 2.4). For systems admitting unlimited refinement no uniform floor exists and the results do not apply as stated.
2. Order independence is graded, not exact (theorem 5.8). We measured the reduction on one family of instances—chains of six items—and the numbers in table 3 are properties of that family. We claim the ordering of the counts, not their values.
3. Theorem 4.7 guarantees invariance only against modifications outside the ball. Commitment reaches inside neighbouring balls, which is exactly why theorem 5.8 is graded, and we have not characterised which mutation disciplines would restore exactness.
4. Powers are derived from a contact graph, not from chemical structure. The graph is itself a model, and the suite records as NOT TESTED the question of whether a derived power matches any measured rate.
5. The ladder inherits the reversal limitation (theorem 6.3): it cannot orient a process,

and orientation remains a question for the medium.

6. Theorem 9.12 establishes conservativity syntactically—no existing rule is modified and added rules apply only to added syntax. It does not establish that no existing program’s *meaning* changes under a reading that mixes the two composition rules, and we have not investigated such readings.
7. The exhaustive minimisation in eq. (3) is exponential in the ball size. This bounds the radius at which the construction is practical and is why the measurements use small graphs. A maximum-flow implementation would remove the limit and we have not written one.

16 Conclusion

Shakespeare had one composition rule, and it was extensive: residue chaining adds the boundary a process commits, is invariant under reordering, and cannot say how much of a process any step accomplishes. We added a second rule that is intensive, and the two are independent: neither aggregate determines the other, so keeping both is not redundancy.

Three things were harder than expected and are reported as they came out. Intensity is not delivered by a local cut key alone—the normaliser must be local too, and the near-miss control is what localised the failure. Order independence is graded rather than exact, because commitment reaches inside neighbouring balls, which is precisely the case the invariance theorem does not cover. And the radius sweep runs in the direction opposite to the one we predicted, for a reason—ball overlap—that we report beside the counts so the direction is explained rather than merely recorded.

The construct earns its place by refusing. A ladder whose declared rungs cannot reach its declared target is rejected before evaluation, the rejection is tight in the sense that it excludes nothing that could have succeeded, and it is non-vacuous in the sense that presentable ladders fall on both sides of it. A construct

that always succeeded would classify nothing, and the framework says so.

Two published claims are falsified by this extension and both are amended in place: there are now two composition rules rather than one, and freeness is a criterion rather than a property of a single privileged rule. The second amendment is the more useful, because theorem 9.5 replaces a licence granted case by case with a test any rule may be put to.

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